

# SPS\_Data607\_Week7\_DC

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## Working with HTML and JSON

Select **three books** on a single subject of your choice. At least **one book must have multiple authors**. For each book, record the **title**, **authors**, and **two to three additional attributes** of your choosing (e.g., publication year, publisher, ISBN, genre, rating).

Using this information, **manually create two data files**:

- An **HTML file** containing a table with the book data (e.g., `books.html`)
- A **JSON file** representing the same data (e.g., `books.json`)

Unless you are already very comfortable with these formats, you are expected to create both files **by hand** to better understand their structure.

Next, write **R code** (using packages of your choice) to:

1. Load the HTML data into an R data frame
2. Load the JSON data into a separate R data frame
3. Compare the two data frames and determine whether they are identical

## Deliverables

- `books.html`
- `books.json`
- R code that loads and compares both data sources

Package your complete solution in a **Quarto (.qmd) file** and **publish it to rpubs.com**.

Note: Publishing will require making your HTML and JSON files **publicly accessible on the web**.

## Approach

I plan to purchase 3 CISSP-related books from Amazon. I will collect detailed information about these books and organize the data in Excel. From the Excel dataset, I will generate an HTML page and export the data as a CSV file. The CSV file will then be converted into JSON format.

<https://www.amazon.com/dp/1394258410>

<https://www.amazon.com/dp/B0D37YPGPZ>

<https://www.amazon.com/dp/1119789990>

## Code base

R-Package rvest, jsonlite, dplyr

```
library(rvest)
library(jsonlite)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
week7url<-"https://github.com/dyc-sps/SPS_DATA607_Week7/raw/5c0c3b23194f0ad491f745999192b638"

htmlpage <- read_html(week7url)
df_html <- html_table(htmlpage,header = TRUE) # after used identical function then add this

df_html <- df_html[[1]]
df_html <- df_html[!apply(df_html, 1, function(x) all(is.na(x) | x == "")), ]
```

```

week7url2<-"https://raw.githubusercontent.com/dyc-sps/SPS_DATA607_Week7/5c0c3b23194f0ad491f74
df_json<-fromJSON(week7url2)
df_json[df_json == ""] <- NA
df_json

```

```

1          The Official (ISC)2 CISSP CBK Reference (Cissp: Certified Inform
2
3 ISC2 CISSP Certified Information Systems Security Professional Official Study Guide & Pract
      Author1      Author2      Author3      Author4
1 Arthur J. Deane      Aaron Kraus      <NA>      <NA>
2   Rob Witcher      John Berti      Lou Hablas Nick Mitropoulos
3   Mike Chapple James Michael Stewart Darril Gibson      David Seidl
      ISBN-13      publisher publicaton year      ISBN-10
1 978-1119789994      Sybex      2021 1119789990
2 979-8987407721 Destination Certification Inc.      2024      <NA>
3 978-1394258413      Sybex      2024 1394258410
      Print length Edition Rank
1   672 pages      6th 91065
2   532 pages      <NA> 18847
3      <NA>      4th 11112

```

```

identical(df_html,df_json)

```

```
[1] FALSE
```

After reviewing the HTML output, we found that the first row should be set as the header row, and the first column needs to be cleaned to remove the extra (`|r|n`) characters.

```

#library(stringr)
df_html[[1]] <- trimws(gsub("\\r\\n ", "", df_html[[1]]))
#df_html <- lapply(df_html, function(x) gsub("\\r\\n", "", x))
#df_html[] <- lapply(df_html, str_replace_all, "\\r\\n", "")
df_html[2,11] <-NA
df_html[1,4] <-NA
df_html[df_html == ""] <- NA
df_html <- as.data.frame(df_html)
df_html

```

```

1           The Official (ISC)2 CISSP CBK Reference (Cissp: Certified Inform
2
3 ISC2 CISSP Certified Information Systems Security Professional Official Study Guide & Prac
      Author1      Author2      Author3      Author4
1 Arthur J. Deane      Aaron Kraus      <NA>      <NA>
2      Rob Witcher      John Berti      Lou Hablas Nick Mitropoulos
3      Mike Chapple James Michael Stewart Darril Gibson      David Seidl
      ISBN-13      publisher publicaton year      ISBN-10
1 978-1119789994      Sybex      2021 1119789990
2 979-8987407721 Destination Certification Inc.      2024      NA
3 978-1394258413      Sybex      2024 1394258410
      Print length Edition      Rank
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2      532 pages      <NA> 18847
3      <NA>      4th 11112

```

Change all values to text and check if the data matches exactly using the `identical()` function.

```

library(dplyr)

df_html <- df_html %>% mutate(across(everything(), as.character))
df_json <- df_json %>% mutate(across(everything(), as.character))

identical(df_html,df_json)

```

```
[1] TRUE
```

## Conclusion:

After processing the HTML table, we successfully cleaned and prepared the data for analysis. The first row was converted to column headers, all columns were converted to text to ensure consistency, and unwanted line breaks (`\r \n`) were removed. Any empty cells were replaced with NA, and fully empty rows were removed. Finally, a verification using the `identical()` function confirmed that the cleaned data exactly matches the expected format, ensuring the table is accurate and ready for further use.